Atoms and Molecules

The big idea

More than 2,000 years ago, Indian philosopher **Maharishi Kanad** suggested that matter is made of tiny indivisible particles (paramanu). In 1808, **John Dalton** turned this into a scientific theory supported by the laws of chemical combination.

Laws of chemical combination

TWO LAWS

Law of conservation of mass (Lavoisier): mass is neither created nor destroyed in a chemical reaction.
Law of constant proportions (Proust): in a chemical compound, elements are always present in a **definite proportion by mass**. Water always has hydrogen and oxygen in the mass ratio 1 : 8.

Dalton's atomic theory explained these laws: matter is made of atoms; atoms of an element are identical in mass and properties; atoms combine in small whole-number ratios; atoms are neither created nor destroyed in reactions.

Atoms and atomic mass

Atoms are incredibly small (radius about 10^{-10} m). Symbols come from English or Latin names: H, O, C, Na (natrium), Fe (ferrum), K (kalium).

Atomic mass unit (u): 1/12 of the mass of one carbon-12 atom. Atomic masses: H = 1 u, C = 12 u, N = 14 u, O = 16 u, Na = 23 u, Cl = 35.5 u.

Molecules and ions

A **molecule** is the smallest particle of a substance that can exist independently. **Atomicity** = number of atoms in a molecule: monatomic (He, Ar), diatomic (O_2, H_2, Cl_2), triatomic (O_3), polyatomic (P_4, S_8).

Ions are charged particles: cations (+) and anions (–). **Polyatomic ions** act as a unit: NH_4^+, OH^-, NO_3^-, CO_3^{2-}, SO_4^{2-}, PO_4^{3-}.

Writing chemical formulae

CRISS-CROSS THE VALENCIES

Write the symbols, put the valencies below, cross them over and simplify. Aluminium oxide: Al (3), O (2) → **Al_2O_3**. Calcium hydroxide: Ca (2), OH (1) → **$Ca(OH)_2$**.

Molecular mass and the mole

Molecular mass = sum of atomic masses. H_2O = 2(1) + 16 = **18 u**.

THE MOLE

One **mole** of any substance contains **6.022×10^{23}** particles (Avogadro's number, N_0).
The mass of one mole in grams equals its atomic/molecular mass in u: 1 mole of water = 18 g.

MOLE CALCULATIONS

number of moles n = given mass m / molar mass M = number of particles N / N_0

HOW MANY MOLES ARE IN 52 G OF HELIUM (ATOMIC MASS 4 U)?

n = 52/4 = **13 mol**.

Quick recap

- Conservation of mass and constant proportions.
- Atomic mass in u; molecular mass = sum of atomic masses.
- Criss-cross valencies to write formulae.
- 1 mole = 6.022×10^{23} particles = molar mass in grams.

Practice questions

1. The law of constant proportions was given by
   (a) Dalton (b) Proust (c) Lavoisier (d) Avogadro
2. The molecular mass of CO_2 is
   (a) 28 u (b) 44 u (c) 32 u (d) 12 u
3. The formula of aluminium oxide is
   (a) AlO (b) Al_3O_2 (c) Al_2O_3 (d) AlO_3
4. The number of particles in one mole is
   (a) 6.022×10^{23} (b) 6.022×10^{22} (c) 1.6×10^{-19} (d) 3×10^{8}
5. 36 g of water contains how many moles?
   (a) 1 (b) 2 (c) 18 (d) 36

Answers: 1. b 2. b 3. c 4. a 5. b